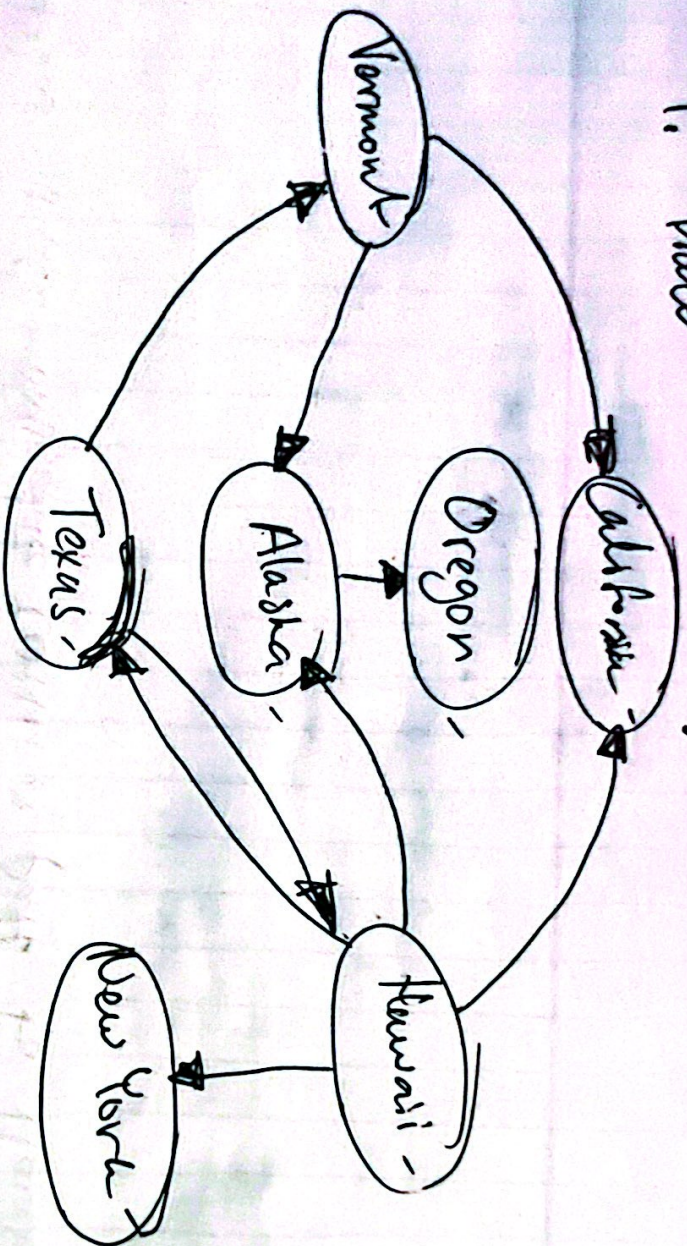


1. Draw the State graph



1. Describe graph pictured above

$V(\text{State graph}) = \{ \text{Alaska, Oregon, Hawaii, Texas, New York, Vermont, California} \}$

$E(\text{State graph}) = \{ (\text{Alaska, Oregon}), (\text{Hawaii, Alaska}), (\text{Hawaii, New York}),$

$(\text{Hawaii, Texas}), (\text{Texas, Hawaii}), (\text{Hawaii, California}),$

$(\text{Texas, Vermont}), (\text{Texas, New York}), (\text{Vermont, California}), (\text{Vermont, Alaska})$

2. a) is there a path from Oregon to any other state in the graph?

No, there is only a path from Alaska to Oregon and not a path from Oregon to any other place.

b) is there a path from Hawaii to every other state in the graph?

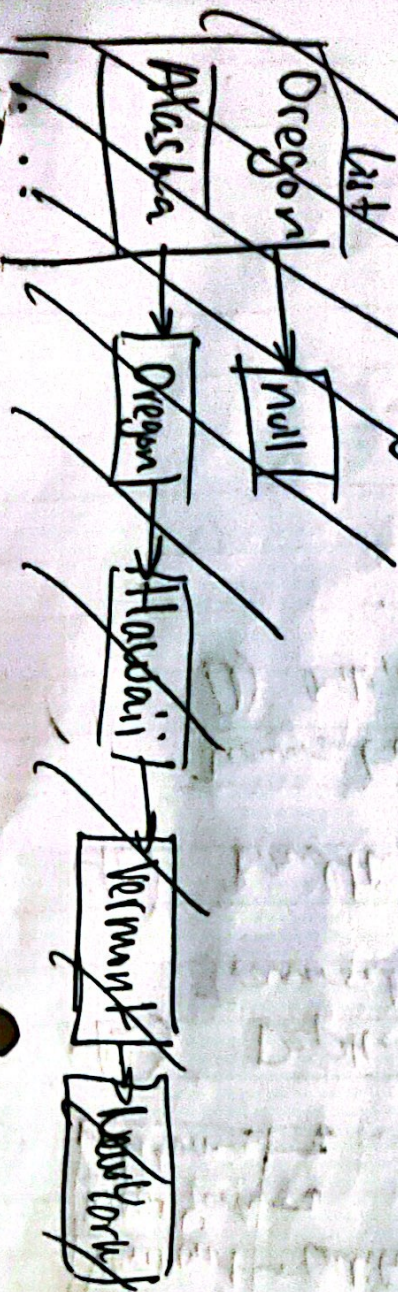
No, Hawaii does not have a path to Oregon or Vermont.

c) Texas is the only state with a path to Hawaii.

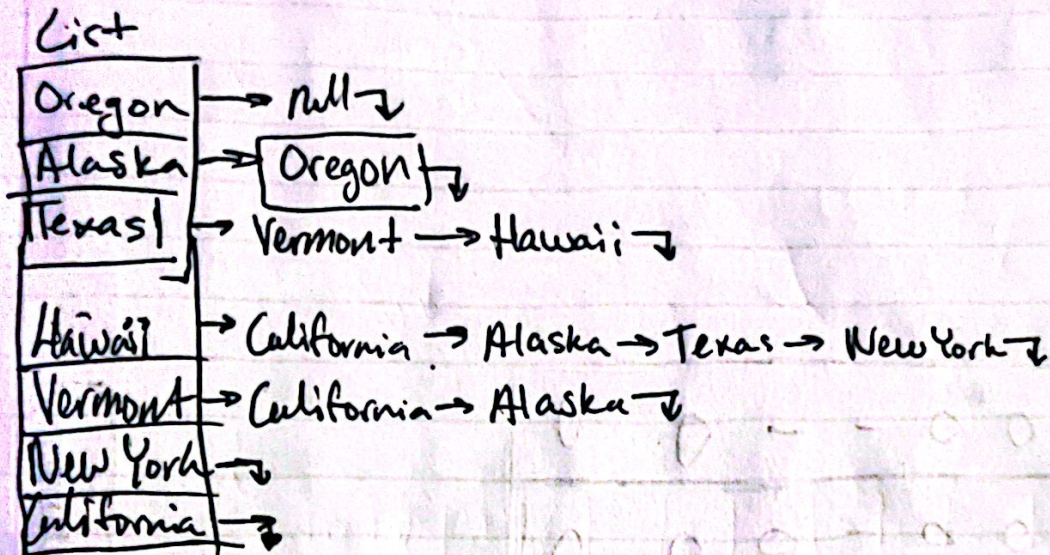
3) adjacency Matrix

States		0	1	2	3	4	5	6
0	Oregon	0	1	0	0	0	0	0
1	Alaska	1	0	0	1	0	0	0
2	Texas	0	0	0	1	0	0	0
3	Hawaii	0	1	1	0	1	1	1
4	Vermont	0	1	1	0	0	0	1
5	New York	0	0	1	0	0	0	0
6	California	0	0	0	1	0	0	0

b) Adjacency list



3b) Adjacency list



4a) Depth First Order at E
 Traversal order: E, G, A, D, F, C, B

4b) Breadth first order at F

Traversal order: ~~E, G, A, D, F, C, B~~

D) a, b, and c are all Breadth first traversals

5. Find Shortest distance from ~~Atlanta~~ to every other city

Atlanta-Washington; Atlanta $\xrightarrow{600}$ Washington
Atlanta-Houston; Atlanta $\xrightarrow{800}$ Houston

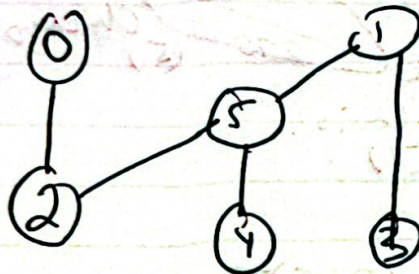
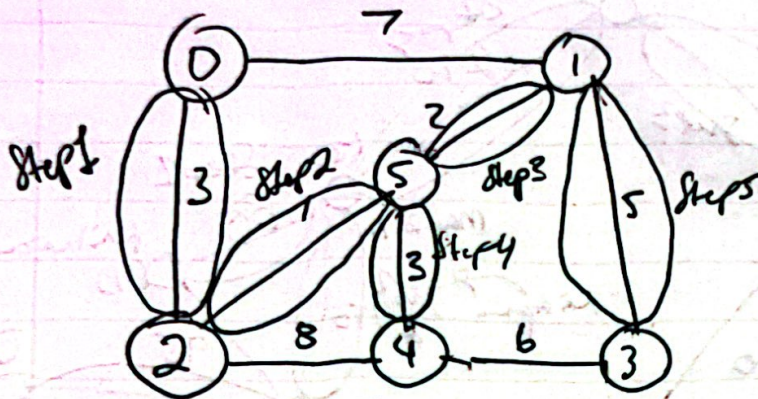
Atlanta-Dallas; Atlanta $\xrightarrow{600}$ Washington $\xrightarrow{1300}$ Dallas

Atlanta-Denver; Atlanta $\xrightarrow{600}$ ~~Atlanta~~ Washington $\xrightarrow{1300}$ Dallas $\xrightarrow{700}$ Denver

Atlanta-Chicago; Atlanta $\xrightarrow{600}$ Washington $\xrightarrow{1300}$ Dallas $\xrightarrow{900}$ Chicago

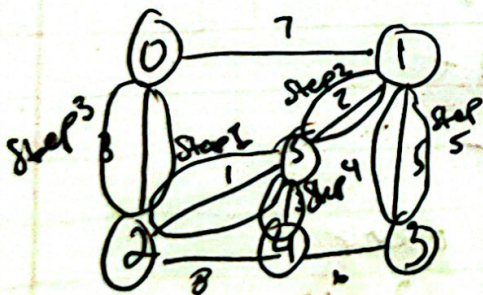
Atlanta-Austin; Atlanta $\xrightarrow{600}$ Washington $\xrightarrow{1300}$ Dallas $\xrightarrow{200}$ Austin

6) Prim's
Start at Zero, $\emptyset$



Edge weight
 $= 3 + 1 + 2 + 3 + 5$
 $= 14$

7) Kruskal's algorithm
 Show weights in order and steps



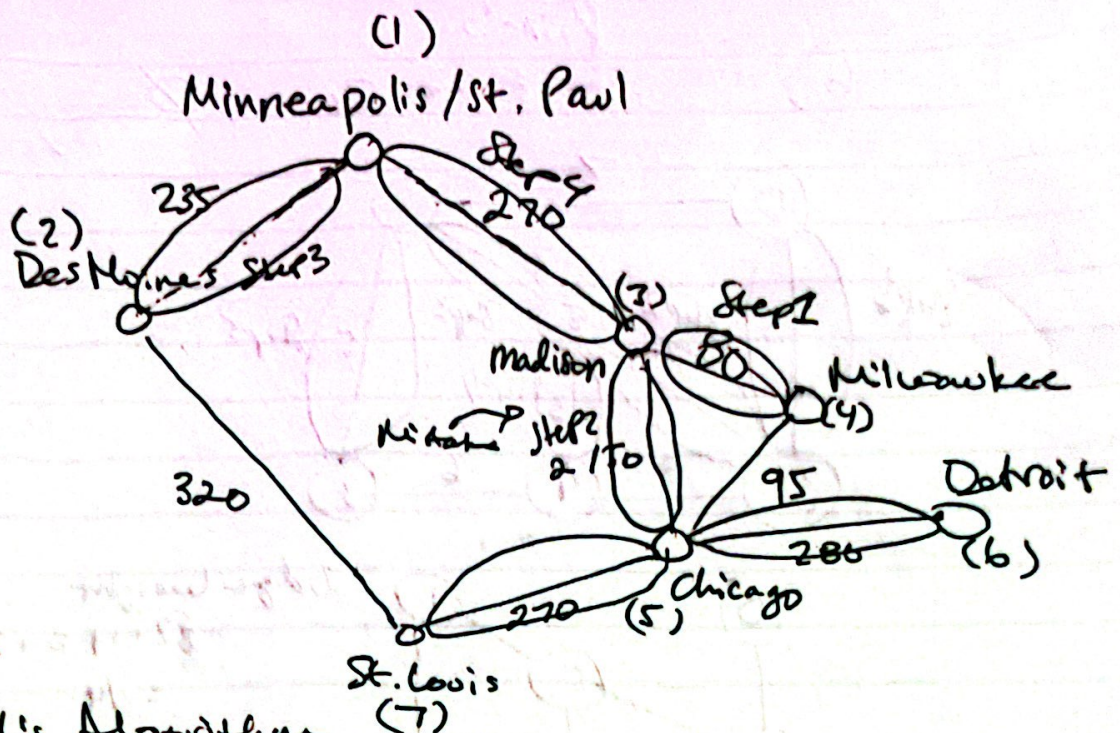
0-1(7)-
 0-2(3)-
 1-5(2)-
 1-3(5)-
 2-5(1)-
 2-4(8)-
 3-4(6)-
 4-5(3)-

ordered by weight

2-5(1)-
 1-5(2)-
 0-2(3)-
 4-5(3)-
 1-3(5)-
 3-4(6) X
 0-1(7) X
 2-4(8) X

} not needed

8)



Kruskal's Algorithm

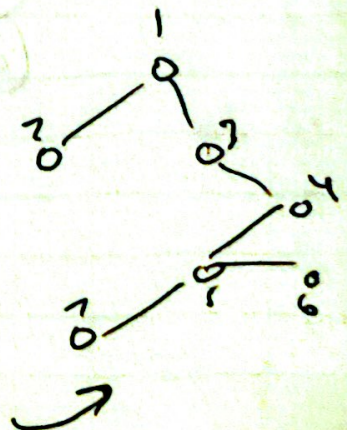
(1) - (2) M/St. Paul - Des Moines (235) -
 (1) - (3) M/St. Paul - Madison (270) -
 (2) - (7) Des Moines - St. Louis (320) -
 (3) - (4) Madison - Milwaukee (80) -
 (3) - (5) Madison - Chicago (150) -
 (5) - (6) Chicago - Detroit (286) -
 (5) - (7) Chicago - St. Louis (270) -

3-4(80) -
 3-5(150) *
 1-2(235) -
 1-3(270) -
 5-7(270) -
 5-6(286) -
 2-7(320) x

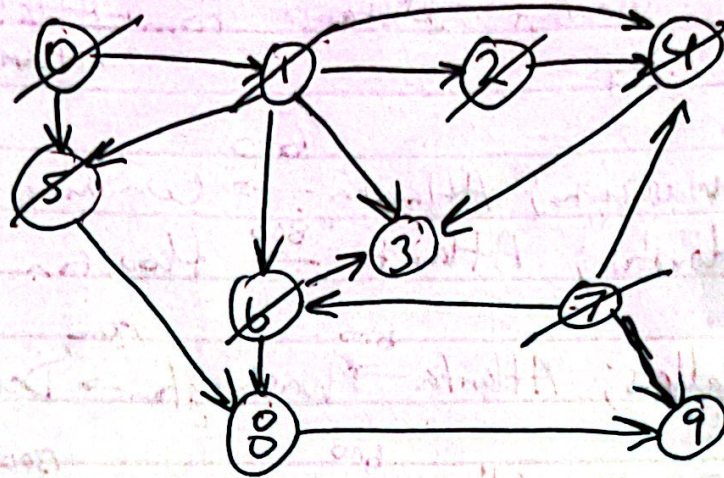
Actual

order

3-4(80) -
 4-5(95) -
 3-5(150) x
 1-2(235) -
 1-3(270) -
 5-7(270) -
 5-6(286) *
 2-7(320) x



Topological Order: 0, 7, 1, 2, 5, 6, 4, 8, 3, 9



Pred Count:

0	1	2	3	4	5	6	7	8	9
0	1	1	3	3	2	2	0	2	2

remove 0: 0 0 1 3 3 1 2 0 2 2

remove 7: 0 0 1 3 2 1 1 0 2 1

~~5~~ remove 1: 0 0 0 2 1 0 0 0 2 1

remove 2: 0 0 0 2 0 0 0 0 2 1

remove 5: 0 0 0 2 0 0 0 0 1 1

remove 6: 0 0 0 1 0 0 0 0 0 1

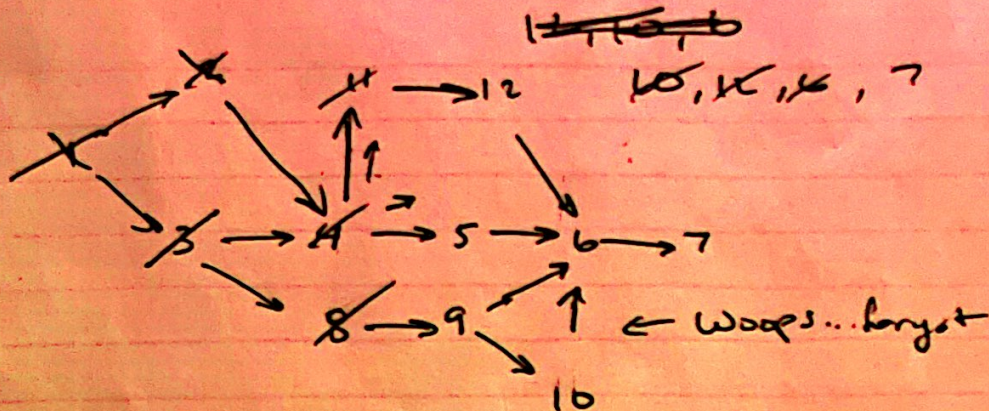
Queue: ~~1~~, ~~2~~, ~~3~~, ~~4~~, ~~5~~, ~~6~~, ~~7~~, ~~8~~, ~~9~~

remove 4: 0 0 0 0 0 0 0 0 1

remove 8: 0 0 0 0 0 0 0 0 0

Top. Order: 1, 2, 3, 4, 8, 5, 11, 9, 10,

Queue: ~~1~~, ~~2~~, 3, 4, 8, 5, 11, 9, ~~10~~



Preal Count:

1	2	3	4	5	6	7	8	9	10	11	12
0	1	1	2	1	3	1	1	1	1	1	1

remove 1: 0, 0, 0, 2, 1, 3, 1, 1, 1, 1, 1, 1

remove 2: 0, 0, 0, 1, 1, 3, 1, 1, 1, 1, 1, 1

remove 3: 0, 0, 0, 0, 1, 3, 1, 0, 1, 1, 1, 1

remove 4: 0, 0, 0, 0, 0, 3, 1, 0, 1, 1, 0, 1
1 2 3 4 5 6 7 8 9 10 11 12

remove 8: 0 0 0 0 0 3 1 0 0 1 0 1

remove 5: 0 0 0 0 0 2 1 0 0 1 0 1

remove 11: 0 0 0 0 0 2 1 0 0 1 0 0

remove 9: 0 0 0 0 0 1 1 0 0 0 0 0

remove 10: 0 0 0 0 0 0 1 0 0

remove 12: takes out 6

remove 6: takes out 7

1, 2, 3, 4, 8, 5, 11, 9, 10, 12, 6, 7

Start, Discrete Math, programming 1, Programming 2,
Computer Organization, High level languages,
Algorithms, Operating Systems, Computers, Theory of Comp.
, Senior Seminar, end